Zeng, Yuxuan

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EDUCATION

Ohio State University — **Columbus, USA**
Doctor of Philosophy, Cognitive Neuroscience in Psychology — 09/2021 - Present

University of Birmingham, Graduate with Distinction — **Birmingham, UK**
Master of Science, Brain Imaging and Cognitive Neuroscience — 10/2018 – 10/2019

Sichuan University — **Chengdu, China**
Bachelor of Science, Biological Sciences — 09/2014 - 07/2018

WORK EXPERIENCES

Research Assistant, Chinese Academy of Science — **Beijing, China**
Advisor: Dr. Li Hu — 08/2020 – 08/2021

Visiting Research Assistant, Hong Kong University of Science and Technology — **Hong Kong, China**
Advisor: Dr. Hyokeun Park — 08/2017 - 01/2018

PUBLICATIONS & PRESENTATIONS & PATENTS

Publications

- **Zeng, Y.,** Hentz, R. E., & Osher, D. E. Dissociable neural coding of configuration and feature information in visual cortex. (in prep)
- **Zeng, Y.,** Hentz, R. E., & Osher, D. E. Post-criterion exposure shapes the development of holistic processing. (in prep)
- **Zeng, Y.**, Lu, Z., Hentz, R. E, & Osher, D. E. (2026). Face-like holistic Processing in Non-Face Stimuli. *Cognitive Psychology.* (in press)
- **Zeng, Y.,** Oechslin, T. S., Widmer, D. E., Kulp, M. T., Fogt, N., Toole, A., ... & Osher, D. E. (2024). Neural consequences of symptomatic convergence insufficiency: A small sample study. *Ophthalmic and Physiological Optics*, *44*(3), 537-545.
- Li, Z., Zhang, L., **Zeng, Y.,** Zhao, Q., & Hu, L. (2023). Gamma-Band Oscillations of Pain and Nociception: A systematic review and meta-analysis of human and rodent studies. *Neuroscience & Biobehavioral Reviews*, 105062. https://doi.org/10.1016/j.neubiorev.2023.105062
- Wang, F., Zhang, L., Yue, L., **Zeng, Y.**, Zhao, Q., Gong, Q., ... & Hu, L. (2021). A novel method to simultaneously record spinal cord electrophysiology and electroencephalography signals. *NeuroImage*, *232*, 117892. https://doi.org/10.1016/j.neuroimage.2021.117892
- Wang, X. Q., Mokhtari, T., **Zeng, Y. X.,** Yue, L. P., & Hu, L. (2021). The Distinct Functions of Dopaminergic Receptors on Pain Modulation: A Narrative Review. *Neural Plasticity*. https://doi.org/10.1155/2021/6682275

Presentations

- **Zeng, Y.**, Hentz, R. E., & Osher, D. E. (2026). Holistic processing requires expertise, but expertise may develop rapidly. *Vision Science Society 2026.* (Conference)
- **Zeng, Y.**, Hentz, R. E., & Osher, D. E. (2025). Rethinking Configurational and Feature-Based Processing in Visual Recognition. *Vision Science Society 2025.* (Conference)
- **Zeng, Y.**, Hentz, R. E., & Osher, D. E. (2024). Selectivity to object features and configuration in the ventral visual stream. *Organization for Human Brain Mapping 2024.* (Conference)
- **Zeng, Y.**, Hentz, R. E., & Osher, D. E. (2024). An Occipitotemporal Region that Identifies Relevant Features. *Vision Science Society 2024.* (Conference)
- **Zeng, Y.,** Fogt N., Kulp M., Toole, A., ... & Osher, D.E. (2023). The Neural Consequences of Symptomatic

Convergence Insufficiency and Reading. *Organization for Human Brain Mapping 2023.* (Conference)

- **Zeng, Y.**, Hentz, R. E., Osher, D. E. (2023). A novel framework to study configural and holistic processing. *Vision Science Society 2023*. (Conference)

Patents

- An Adaptive Multimodal Emotion Regulation System and Method, 2021, CN112618913A
- A System for Synchronous Recording of Brain and Spinal Cord Electrophysiological Signals, 2021, CN215272834U
- A System and Method for Synchronous Recording of Brain and Spinal Cord Electrophysiological Signals, 2021, CN112754499A

RESEARCH EXPERIENCES

Graduate Student, the Ohio State University — **Columbus, USA**

Configuration and feature representations in humans, models, and the brain — 09/2021 - Present

Advisor: Dr. David Osher

- Designed controlled visual learning paradigms to dissociate relational configuration from local feature information, enabling precise tests of how different information types shape recognition in humans and artificial neural networks.
- Led a published behavioral study showing that non-face stimuli can elicit face-like holistic processing after brief configuration-based learning, and compared human behavioral signatures with convolutional neural networks trained under matched configural versus featural regimes.
- Built and analyzed CNN training pipelines to test whether relationally trained models develop human-like vulnerabilities to inversion, misalignment, composite, and part-based manipulations, revealing convergent sensitivity to spatial structure across biological and artificial systems.
- Developed an fMRI project examining how configuration and feature information are differentially represented across dorsal and ventral visual cortex, including category-selective regions, using ROI-based analyses, multirun preprocessing pipelines, and model-driven hypotheses about neural coding.
- Designed an ongoing behavioral study to characterize post-criterion learning and the emergence of holistic processing across multiple task markers, testing whether exposure alone or the content of learned information drives expert-like processing.
- Conduct quantitative analyses in Python and MATLAB for behavioral, neural, and modeling data, including preprocessing, statistical testing, visualization, and reproducible analysis workflows across large experimental datasets.

Brain connection and reading deficit in convergence insufficiency

- Processed and analysis DTI data of subjects with convergence insufficiency and normal binocular vision, and prepared manuscript of related publication

Master Student, Department of Psychology, University of Birmingham — **Birmingham, UK**

Real-world objects brain representation comparison during visual, auditory and audio-visual processing — 10/2018 - 10/2019

Advisor: Dr. Ian Charest & Dr. Jasper van den Bosch

- Created and standardized 4 categories (human, animal, natural, and artificial) of real-world video stimuli
- Used above videos (audio-visual) along with sounds (auditory) and images (visual) extracted from the videos to induce brain responses for semantically similar contents
- Constructed representational dissimilarity matrices (RDMs) by comparing participants' behavior (obtained from a multi-arrangement task) and related EEG data across modalities in the whole brain
- Found preliminary evidence for good category clustering across all three modalities

TEACHING EXPERIENCES

Course Assistant

Data Analysis in Psychology, Introduction to Cognitive Neuroscience, Memory & Cognition

Lecture

Sensation & Perception (guest lecture)

INVITED TALKS

The Ohio State University
"Uncovering Configuration (and Feature) Processing in Human Vision." 01/31/2025
The Ohio State University
"Configural processing beyond face perception." 04/12/2024
The Ohio State University
"CONFIGURATION: beyond face perception." 03/31/2023

AWARDS & HONORS

- Graduate Student Conference Presentation Award 2022-2026
- University Fellowship, Ohio State University 2021-2022
- Master of Science graduated with distinction, University of Birmingham 2019
- National Top-Notch Students Fellowship 2014-2018

SERVICES

- CCBBI student group member, 2021-current
- Crisis Text Line volunteer, 08/2022-08/2023
- Active listener in mental support group, 06/2022-09/2023

MENTORSHIPS

Undergraduates

- Ren Hentz
- Maggie Duffie